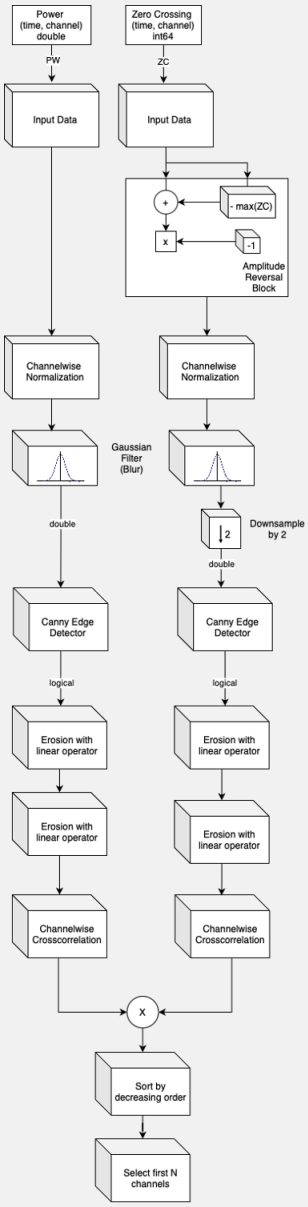
{% include base_path %}

Industry Experience

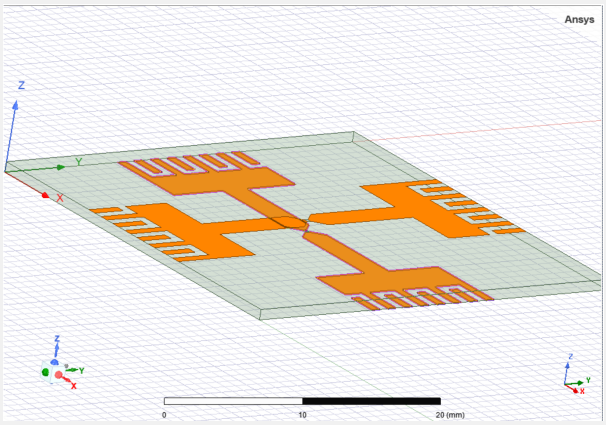
2021 April - 2022 June | Part-Time Signal Processing Engineer in NANOTAM

During my work on processing OTDR data, I utilized classical and efficient signal processing techniques with MATLAB, Python, and C++. I researched the use of chirped pulses as the probing signal for phi-OTDR, calculated the noise profile of the ADC and DAC modules in various projects, and programmed an Nvidia Jetson board for specific computer vision applications. Additionally, I designed a real-time signal processing pipeline in MATLAB for phi-OTDR event detection.

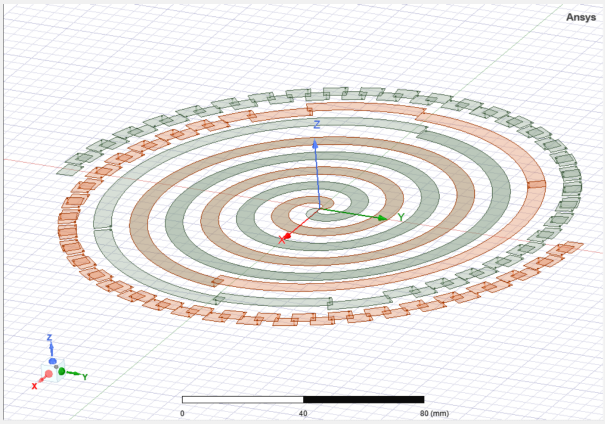


2022 July - August | Summer Internship in ASELSAN

During my internship in the Antenna Design Group, I successfully designed wideband antennas that met specific requirements using Ansys HFSS software. My responsibilities included optimizing the physical dimensions and selecting appropriate superstrate materials for an array antenna.

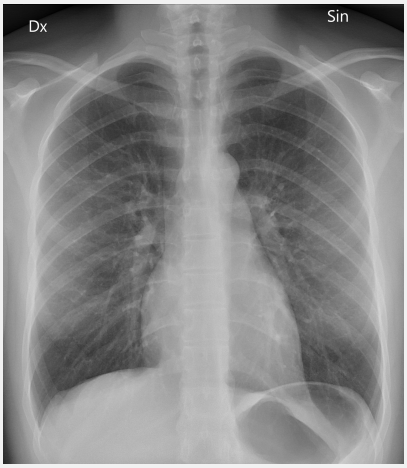


Furthermore, I designed a meandered spiral Archimedean antenna tailored for a specified frequency range, ensuring optimal performance and efficiency.



2021 June - July | Summer Internship in UMRAM

During my summer internship at UMRAM, I focused on several key projects. I conducted extensive research on deep learning models for computer vision, designed medical X-ray image classifiers using transfer learning techniques with backbone models like EfficientNet, ResNet, and Inception, and developed a robust classifier model for detecting monkeypox disease from image data.



Research Experience

2024 Sep - Present | Undergraduate Researcher in Biomedical Imaging Group

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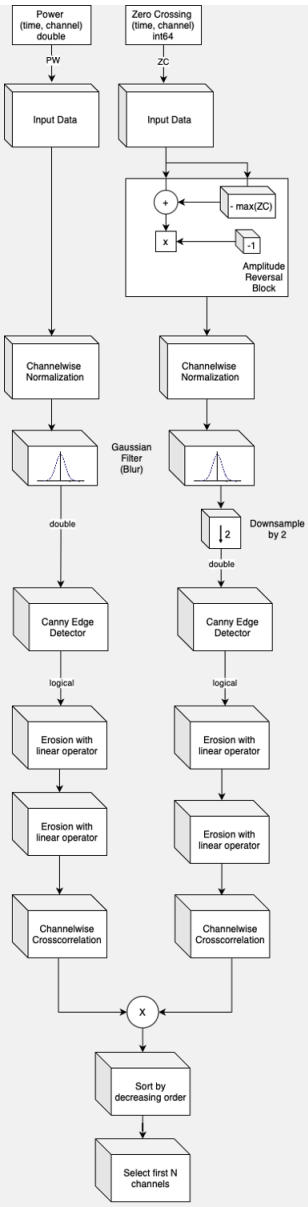
EPFL Image

I have conducted research in BIG Lab



2023 Mar - 2024 Aug | Undergraduate Researcher under Prof. Sinan Gezici

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Bilkent Image

I have conducted research on visible light positioning and the potential use of intelligent reflective surfaces under the supervision of **Prof. Sinan Gezici**. As part of my studies, I completed a semester project titled **ML-Enhanced VLP in IRS-Assisted Indoor Environments with a Single LED Transmitter**. Building upon this work, I authored and published a journal paper, [**IRS Aided Visible Light Positioning with a Single LED Transmitter**] (https://doi.org/10.1016/j.dsp.2024.104799), in *Elsevier Digital Signal Processing*. The study focuses on enhancing localization performance, measured by the Cramér-Rao Lower Bound (CRLB), by leveraging intelligent reflective arrays with a single LED transmitter. ### 2021 Mar - Aug | Undergraduate Researcher in UMRAM

I have worked on Deep Learning on a special focus of biomedical images in the ICON Lab under the Supervision of **Prof. Tolga Çukur**. I have developed classification models for chest tomography image detection tasks and conducted theoretical research on Capsule Networks. The project "Disease Detection from X-Ray Images with Visual Transformer Neural Networks (118C543)", I've worked on, was supported by the **2247 - C STAR-Intern Researcher** Programme.

Coding Skills